Binglin Zhao

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SUMMARY

Research-oriented HCI candidate with a background in industrial design and cognitive experimentation. My work combines controlled experiments, eye-tracking analysis, and interactive system development to investigate how visual and interaction design factors affect cognitive load and decision-making.

EDUCATION

2023.09 – 2025.06	**Master of Engineering (Designology)**	Harbin Institute of Technology
2019.09 – 2023.06	**Bachelor of Engineering (Industrial Design)**	Hebei University of Technology

RESEARCH EXPERIENCE

Research Assistant in Cognitive Neuroscience — Sep 2025 - Dec 2025
Beijing Institute for Brain Research — Advisor: Dr.Qianqian Yu

- Processed and analyzed eye-tracking data from 100+ participants using Python; performed statistical analyses (t-tests, ANOVA) and generated visualizations
- Conducted behavioral and cognitive experiments with clinical populations, including experiment administration, real-time control of experimental flow, and accommodate variable mental conditions
- Resolved technical issues in PsychoPy by debugging and restructuring code to ensure reliable event marker transmission to eye-tracking systems

RESEARCH OUTPUT

Spine Robot Interface Research for Cognitive Load Reduction — Link to Writing Sample

- Conducted a factorial experimental study grounded in Cognitive Load Theory (CLT) to systematically investigate the effects of key interface design factors on cognitive load in spinal surgery robot interfaces.
- Derived a set of evidence-based interface design principles by synthesizing experimental findings from Experiment 1, translating CLT into actionable visual design strategies.
- Designed and conducted a comparative evaluation study (Experiment 2) to assess the effectiveness of the optimized interface against a baseline system, and demonstrated that the optimized interface significantly reduced cognitive load by 39.9% ($p=0.025$) and improved task performance by 16.9% ($p=0.048$)

3D Interactive Experimental System for Episodic Memory Bias Research — Link to Demo

- Translated episodic memory bias theory into a structured experimental workflow and implemented it as a participant-centered 3D interactive paradigm to improve engagement and accessibility
- Developed custom UI and experimental control with real-time data logging, supporting integration with physiological recording devices (iEEG)

SKILLS

Programming	Python, C#	Data Analysis	SPSS, Python
Interactive Dev	Unity, Figma, Adobe Suite	3D Visualization	Rhino, Blender, KeyShot
Other	PsychoPy, LaTeX, Tobii		

Last updated: April 17, 2026